

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech.

Semester : IV

Branch : Mechanical Engineering / Mechanical Engineering (Sandwich)

Subject Code & Name: BTMC402 & Theory of Machines - I

Max Marks: 60

Date: 21/12/2024

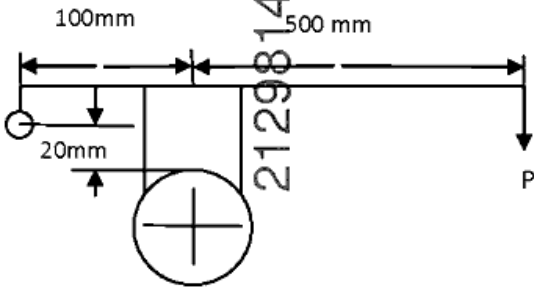
Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per POBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The motion transmitted between the teeth of gears in mesh is a. Sliding b. Rolling c. Turning d. Partly sliding and partly rolling	1/1	1
2	The mechanism forms a structure, when the number of degree of freedom is equal to a. Zero b. One c. Two d. Three	1/1	1
3	In reciprocating steam engine, which of the following forms a kinematic link? a. Cylinder & piston b. Piston rod & connecting rod c. Crank shaft & Fly wheel d. Fly wheel & engine frame	1/1	1
4	The total numbers of instantaneous centers of a mechanism consisting of N links are: a. $N/2$ b. $(N-1)/2$ c. $(N-2)/2$ d. $N(N-1)/2$	1/1	1
5	The locus of instantaneous centre in space during a definite motion of the body is called a. Axode b. Space Centrode c. Body Centrode d. Centro	1/1	1
6	The component of the acceleration, parallel to the velocity of particle at the given instant is called a. Radial b. Tangential c. Coriolis d. Centripetal	1/1	1
7	The Coriolis component of acceleration is taken into account for a. Slider crank b. Four bar c. Quick return d. Geneva	1/2	1

	mechanism	mechanism	mechanism	Mechanism		
8	The angle of inclination of plane, at which the body begins to move down the plane, is called				1/2	1
	a. Angle of friction	b. Angle of response	c. Angle of projection	d. Angle of taper		
9	The friction experienced between two dry and un-lubricated surface in contact is known as				1/3	1
	a. Solid friction	b. Fluid friction	c. Boundary friction	d. Film friction		
10	The ratio of limiting friction to the normal reaction between the two bodies is known as				1/3	1
	a. Coefficient of friction	b. Angle of friction	c. Force of friction	d. Resultant friction		
11	The Bearing and Journal forms				1/4	1
	a. Sliding pair	b. Turning pair	c. Screw pair	d. Spherical pair		
12	A dynamometer is a brake but in addition it has a device to measure				1/4	1
	a. Frictional Resistance	b. Weight	c. Speed	d. RPM		
Q. 2	Solve the following.					12
A)	Write the classification of Kinematic pairs according to the type of contact between the elements.				1/1	6
B)	In a slider crank mechanism the radius of crank and length of connecting rod are 150 mm and 600 mm respectively. If the crank rotates clockwise with the angular velocity of 12 rad/s and is at 45 degree from inner dead center. Find velocity of slider and angular velocity of connecting rod using Klien's construction method.				2/1	6
Q.3	Solve the following.					12
A)	Derive an expression for torque transmitted by single plate clutch due to friction considering uniform pressure theory.				2/2	6
B)	An automobile single plate clutch consists of a pair of contacting surfaces the inner and outer radii of friction plate are 120mm and 250 mm respectively. The coefficient of friction is 0.25 and total axial force				2/2	6

	is 15 KN. calculate the power transmitting capacity of clutch at 500 RPM by using uniform wear theory		
Q. 4	Solve Any Two of the following.		12
A)	Derive an expression for braking torque for single block brake when the line of action of braking force passes through the fulcrum of the lever.	2/3	6
B)	In a rope brake dynamometer diameter of flywheel 1m, diameter of rope 10 mm, dead weight of the brake 50Kg, speed of Engine 180 rpm, spring balance reading 120 N, Find brake power of the Engine.	2/3	6
C)	A single block brake has drum of 2m diameter. It is required braking torque of 500 Nm at 300 rpm clockwise. The coefficient of friction is 0.25. Determine the force P to be applied at the end of lever when angle of contact is 30°. 	2/3	6
Q.5	Solve Any Two of the following.		12
A)	Explain with neat sketch types of cam.	1/4	6
B)	A cam rotating at uniform speed required to give the Knife edge follower motions as: (i) Follower to move outwards through the distance of 30mm during 90° of cam rotation with uniform velocity (ii) Follower to dwell for next 60° of cam rotation (iii) Follower to return to its original position during 90° cam rotation with uniform velocity (iv) Follower to dwell for remaining period. The minimum radius of the cam is 50 mm. Draw the cam profile when follower is offset by 20mm from the axis of the cam shaft	3/4	6
C)	Draw the profile of cam to raise valve with S.H.M. through 40 mm in $1/4^{\text{th}}$ of revolution keep it fully raised through $1/10^{\text{th}}$ revolution and to lower it with uniform acceleration and retardation in $1/6^{\text{th}}$ revolution. The valve remains closed during	3/4	6

	rest of the revolution. The diameter of the roll is 20 mm and minimum radius of cam to be 30 mm. The axis of the valve rod passes through the axis of cam shaft. The cam shaft rotates at 360 rpm clockwise. Draw the cam profile when follower translates along the axis of the cam.		
Q. 6	Solve Any Two of the following.		12
A)	Explain the term 'static balancing' and 'dynamic balancing'. State the necessary conditions to achieve them.	2/5	6
B)	Four masses m_1 , m_2 , m_3 and m_4 are 200 kg, 300 kg, 240 kg and 260 kg respectively. The corresponding radii of rotation are 0.2 m, 0.15 m, 0.25 m and 0.3 m respectively and the angles between successive masses are 45° , 75° and 135° . Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2 m using analytical method.	3/5	6
C)	A, B, C and D are four masses carried by a rotating shaft at radii 100, 125, 200 and 150 mm respectively. The planes in which the masses revolve are spaced 600 mm apart and the mass of B, C and D are 10 kg, 5 kg, and 4 kg respectively. Find the required mass A and the relative angular settings of the four masses so that the shaft shall be in complete balance	3/5	6
*** End ***			